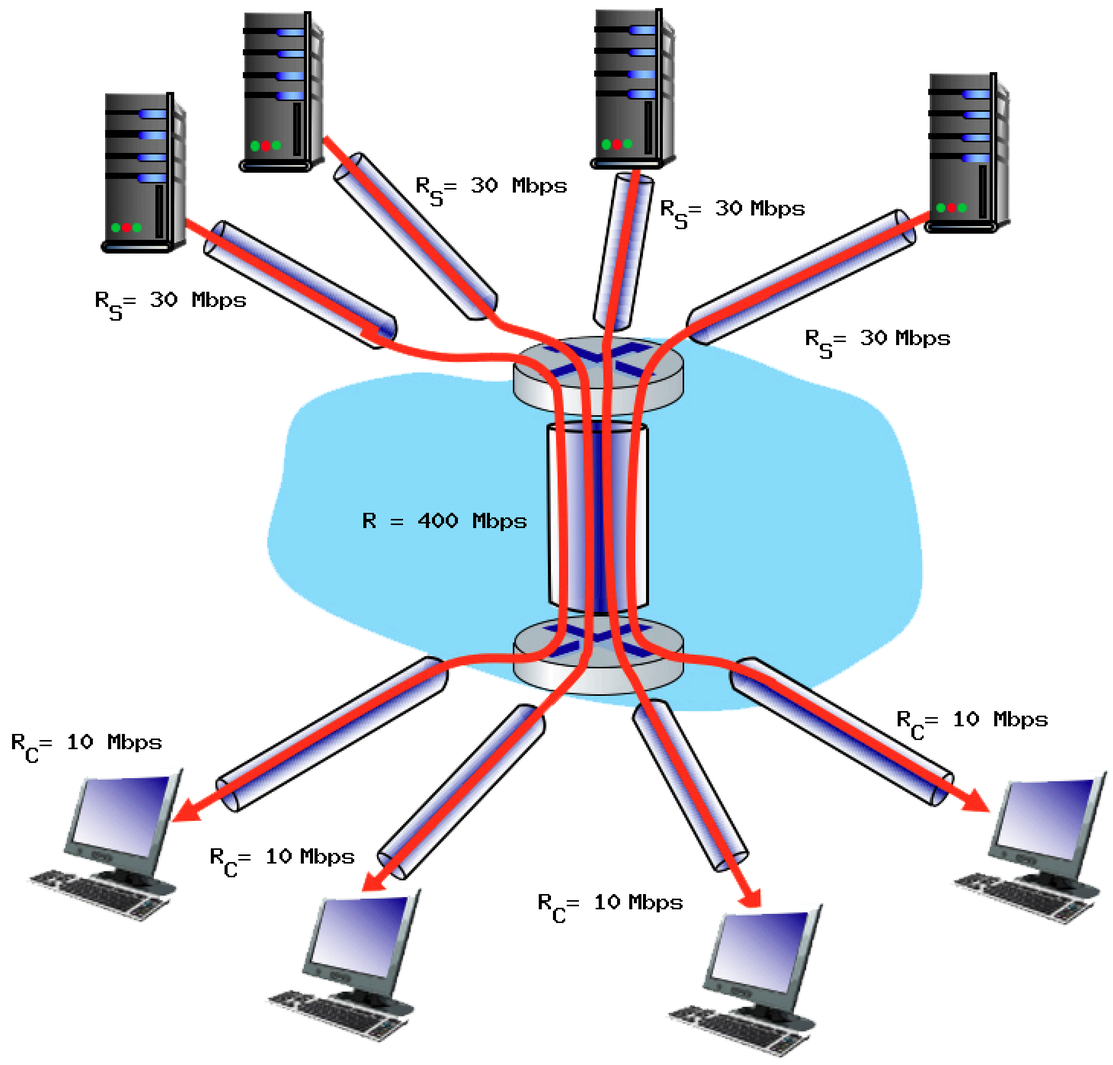
INTERACTIVE END-OF-CHAPTER EXERCISES

END TO END THROUGHPUT AND BOTTLENECK LINKS

Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of R = 400 Mbps. The four links from the servers to the shared link have a transmission capacity of R_S = 30 Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of R_C = 10 Mbps.



You might want to review Figure 1.20 in the text before answering the following questions

QUESTION LIST

1. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

2. Which link is the bottleneck link? Format as Rc, Rs, or R

3. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S)? Answer as a decimal

4. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_C)? Answer as a decimal

5. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal

SOLUTION

1. The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 10 Mbps

2. The bottleneck link is the link with the smallest capacity between R_S, R_C, and R/4. The bottleneck link is Rc.

3. The server's utilization = $R_{bottleneck}$ / R_S = 10 / 30 = 0.33

4. The client's utilization = $R_{bottleneck}$ / R_C = 10 / 10 = 1

5. The shared link's utilization = $R_{bottleneck}$ / (R / 4) = 10 / (400 / 4) = 0.1

QUESTION 1 OF 5

What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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